

Date: 30 July 2026

To
Ms Faiza Feroz
Lecturer
Computer Graphics Lab
Daffodil International University.

Subject: Request for Permission to Proceed with **"An Interactive Smart Traffic Management Simulation using OpenGL and C++"**.

Dear Ma'am,

I (Samiul Haque & Affan Julfiker) my friend as a team, would like to propose a new project titled "An Interactive Smart Traffic Management Simulation using OpenGL and C++" for the Computer Graphics Lab under your supervision this semester.

Our project focuses on developing a 2D interactive traffic management simulation using C++, OpenGL, and GLUT. The simulation will model a city intersection with roads, buildings and moving vehicles, day/night view along with an intelligent traffic signal system that manages signal switching, vehicle flow and other real-time interactive elements to create a realistic and engaging visual environment.

Through our this project, we aim to strengthen our understanding of computer graphics concepts including 2D drawing, transformations, animation and event handling using OpenGL. Therefore our project outcome will be an educational and visually appealing graphics application that can be further enhanced with advanced features in the future.

We kindly request your permission to proceed with this project. Your guidance and approval will be invaluable for its successful completion.

Thank you for your time and consideration.

Sincerely,

Team Members:	ID
1. Samiul Haque	232-15-098
2. Affan Julfiker	232-15-682

An Interactive Smart Traffic Management Simulation using OpenGL and C++

Project Overview

Urban traffic congestion is a major challenge in modern cities, making efficient traffic signal management essential for improving road safety and reducing traffic delays. This project presents an interactive **2D Smart Traffic Signal Simulation** developed using **C++** and **OpenGL (GLUT)**. The simulation models a city intersection with roads, buildings, vehicles, and traffic lights, while allowing users to control signals, switch between day and night modes, and simulate rainy weather through keyboard and mouse interactions. It demonstrates how computer graphics can be used to visualize real-world traffic management systems in an interactive and educational way.

Problem Statement

Traditional learning methods cannot effectively demonstrate how traffic signals coordinate traffic flow under different environmental conditions.

A graphical simulation provides a better understanding of:

- Traffic signal operations. (R-RED / G-GREEN)
- Vehicle movement.
- Road intersection management.
- Environmental effects on road visibility.
- User interaction with real-time graphics.

Project Objectives

The primary objectives of this project are:

1. Develop an interactive traffic signal simulation.
2. Simulate a four-way city intersection.
3. Control horizontal and vertical traffic signals.
4. Implement realistic Day and Night environments.
5. Add Rain Animation to enhance realism.
6. Demonstrate object animation using OpenGL.
7. Improve understanding of Computer Graphics concepts.
8. Build a user-friendly simulation for educational purposes.

Project Features

Smart Traffic Signal Control

- Red and Green signal switching
- Separate control for horizontal and vertical roads

Dynamic Environment

- Day Mode
- Night Mode

☁ Weather Simulation

- Real-time Rain Animation.
- Rain Start/Stop using mouse controls.

🚗 Animated Vehicles

- Moving cars.
- Smooth vehicle animation.

🏙 City Environment

- Roads
- Buildings
- Trees
- Footpaths
- Walking Area
- Streetlights
- Traffic Lights
- Road Markings

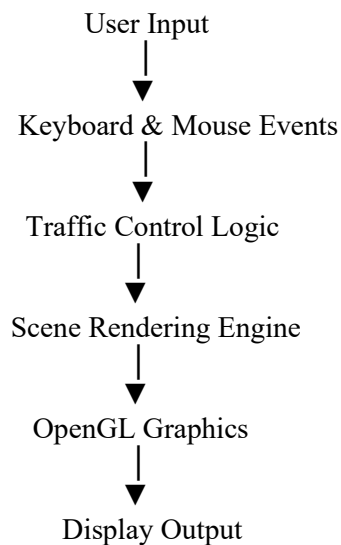
🎮 Interactive Controls

- Keyboard Letter-based traffic control.
- Mouse-based weather control.
- Real-time scene update.

Technology Stack

Component	Technology
Programming Language	C++
Graphics Library	OpenGL
Toolkit	GLUT
IDE	CodeBlocks / Visual Studio
Operating System	Windows

System Architecture



Functional Requirements

Keyboard Controls:

Key	Action
R	Horizontal Road → RED, Vertical Road → GREEN
G	Horizontal Road → GREEN, Vertical Road → RED
D	Day Mode
N	Night Mode

Mouse Controls:

Mouse	Action
Left Click	Start Rain
Right Click	Stop Rain

Modules of the Project

Module 1 – City Environment

- Roads
- Buildings
- Trees
- Footpaths
- Walking Area

Module 2 – Traffic Signal System

- Traffic Lights
- Signal Switching

Module 3 – Vehicle Animation

- Vehicle Movement
- Motion Control

Module 4 – Environment Controller

- Day View
- Night View

Module 5 – Weather Controller

- Rain Animation
- Rain Effects

Module 6 – User Interaction

- Keyboard Events
- Mouse Events

Expected Outcomes

After successful completion, the project will:

- Simulate a realistic city traffic intersection.
- Demonstrate real-time traffic signal operations.
- Show environmental changes dynamically.
- Improve practical knowledge of OpenGL programming.
- Provide an educational visualization of traffic management systems.

Project Benefits

- ☐ Easy visualization of traffic systems.
- ☐ Interactive learning experience.
- ☐ Improves OpenGL programming skills.
- ☐ Demonstrates real-world traffic management concepts.
- ☐ Useful for academic demonstrations and presentations.

Conclusion

The **Traffic Signal in the City** project is an interactive computer graphics application that combines traffic management with environmental simulation. By integrating OpenGL rendering, animation, and user interaction, the project provides a practical demonstration of real-world traffic signal operations in a virtual city environment.

It serves as an educational tool for learning **Computer Graphics, OpenGL programming, event handling, animation, and simulation design**, while also laying the foundation for future smart traffic management systems.

END